

## Lab 2 — Building Version 1 and Publishing to GitHub

After Module 2 — GitHub Copilot Introduction

Open the TodoApp folder from Lab 1 in VS Code. Confirm docs/PRD.md and docs/TRD.md are present, and that Python 3.12, Node 20, and Git are installed.

### Step 1 — Switch to Agent mode

In Copilot Chat, set the mode picker to Agent and the model picker to Claude Sonnet 5.

Agent mode can create files, run terminal commands, and correct its own errors. Watch the chat panel as it works — you will be asked to approve terminal commands.

### Step 2 — Build the application

Type this prompt and press Enter.

Build Version 1 of the Todo application exactly as specified in #file:docs/TRD.md.

Follow the TRD precisely:

- The folder structure in section 2.
- The data model in section 3.
- The API endpoints in section 4, including the status codes.
- The three backend layers in section 5 - routers, services, repository. The repository is the only file that stores data.
- The frontend components in section 6.

Additional constraints:

- Backend: Python 3.12, FastAPI, Pydantic. Create a requirements.txt.
- Store todos in a Python list inside the repository layer only.
- Frontend: React created with Vite, JavaScript (not TypeScript), functional components, useState.
- Enable CORS on the backend so the React dev server can call it.
- Write the code like a junior developer would - simple and readable.

Do NOT include:

- Any database or ORM
- A Category field
- Authentication or authorization
- Exception handling beyond what FastAPI provides by default
- Docker, tests, or CI configuration

When the code is written, create a README.md at the root with the exact commands to run the backend and the frontend.

Approve the terminal commands when prompted. This will take several minutes.

### Step 3 — Run and test the application

Open two terminals in VS Code (Ctrl + Shift + backtick).

Terminal 1 — backend:

```
cd backend
pip install -r requirements.txt
uvicorn main:app --reload
```

Terminal 2 — frontend:

```
cd frontend
npm install
npm run dev
```

Open the URL shown by the frontend terminal, usually <http://localhost:5173>. Also open <http://localhost:8000/docs> to see the FastAPI interactive API documentation.

Add three todos. Mark one complete. Delete one. Refresh the page — the todos are still there.

Now stop the backend terminal with Ctrl + C and start it again. Refresh the browser. The todos are gone. This is Version 1 behaving exactly as the PRD specified. Version 2 will fix it.

#### Step 4 — Fix anything broken

If the application does not run, do not fix it by hand. Select the error in the terminal, then in Copilot Chat:

The application fails to start with this error:

```
#terminalSelection
```

Read the relevant files and fix the cause. Do not change the folder structure defined in `#file:docs/TRD.md`.

#### Step 5 — Create the .gitignore

In Copilot Chat:

Create a .gitignore at the repository root suitable for a project with a Python FastAPI backend and a React Vite frontend in the same repository. Include Python virtual environments, `__pycache__`, `node_modules`, build output, and editor folders.

Open the generated .gitignore and read it. Confirm `node_modules/` and `__pycache__/` are listed.

#### Step 6 — Initialise the repository

In the terminal, from the TodoApp root:

```
git init
git add .
git status
```

Read the output of `git status` carefully. If `node_modules` or `__pycache__` appear in the list, your `.gitignore` is wrong or was added too late. Fix it before continuing.

```
git commit -m "Version 1: Todo app with in-memory storage, built from TRD"
```

### Step 7 — Create the GitHub repository and push

- On [github.com](https://github.com) click + in the top right, then New repository.
- Name it TodoApp. Set it to Private. Do not add a README, `.gitignore`, or licence.
- Click Create repository.

Copy the commands under "push an existing repository" and run them:

```
git remote add origin https://github.com/<your-username>/TodoApp.git
git branch -M main
git push -u origin main
```

Refresh the GitHub page. Confirm `docs/PRD.md`, `docs/TRD.md`, `backend/`, and `frontend/` are all present.

### What you should have at the end of this lab

```
TodoApp/
.gitignore
README.md
docs/PRD.md
docs/TRD.md
backend/
frontend/
```

Running locally and pushed to GitHub.